

Célian DI GIOVANNI

Information

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Antibes

Driver's License B

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celian06.github.io

Skills

- Python: (PyTorch, Numpy, Pandas, TensorFlow, Git) ●●●○○
- Signal processing: ●●●○○
  ECG, sEEG, Bio-impedance
- Biostatistics ●●●○○
- AI & Data Science: ●●●○○
  Deep Learning (CNN, Autoencoder), Machine Learning, Feature Engineering, Dynamic stratification, PCA, Statistical Significance
- Neurosciences ●●○○○
- Computer Vision: ●○○○○
  Segmentation, thresholding
- Java / C ●○○○○

Professional Summary

Graduate Engineer from ISEN Méditerranée specialized in Artificial Intelligence, Signal Processing, and Data Science. Experienced in R&D and clinical research (INSERM) processing massive time-series datasets via Deep Learning, and award-winner for embedded AI development on microcontrollers (STM32).

Seeking a challenging technical opportunity (R&D Engineer, Data Scientist, AI / Edge AI Engineer, or PhD candidate) to design, evaluate, optimize, and deploy complex algorithms for industrial systems.

Education

Master of Science in Digital & Electronic Engineering 2023 - 2026
ISEN Méditerranée, Toulon, France

Bionics Engineering – Erasmus Exchange Semester Sept. 2024 - Feb 2025
Università di Pisa, Italy

CPGE PCSI / PC – Prep School in Math, Physics and Chemistry 2021 - 2023
Lycée International de Valbonne, France

High School Diploma in Mathematics & Physics-Chemistry – Honors 2021
Institution du Mont Saint-Jean, Antibes, France

Professional Experience

R&D Engineering Intern – AI & Signal Processing April 2026 - Oct. 2026
INSERM – Institute of Neurosciences, Marseille, France

- **Context:** Deep Learning automation for artifact filtering and large-scale time-series analysis (**~13 million channels**) for the **F-TRACT project** within the international Brainstim team
- **Data Engineering:** Handled **100,000 multi-channel time series**; developed a dynamic stratification algorithm to standardize and balance datasets
- **Modeling:** Architected **PyTorch 2D CNN and multi-task Autoencoder** models to automatically filter noise and **classify signal validity.**
- **Computing:** Parallelized pipeline execution and trained deep network models on high-performance Linux computing clusters
- **Results and deployment :** Delivered a final model achieving **85% accuracy validated by McNemar testing;** created decision-support UI tools for real-time visualization.

Languages

- French native
- English B2 – Cambridge
- Spanish A2

Hobbies

- Trips, History et culture
- Electric and acoustic guitar
- Cinema and Rock Music
- Judo, Badminton, Running, Paddle
- Volunteer work (Ordre de Malte)

Certifications

- First Aid Certification level 1

Engineering Application Intern – QA & Data Engineering
juin - sept. 2025

Pharmacy Department, CHU de Nice, France

- Conducted functional validation for a mobile app (**MUSE project**), defining use cases.
- Defining **use cases and writing test specifications**.
- Served as the lisaison between user feedback groups and software engineering teams.
- Automated data extraction and cleaning from **unstructured DOCX/XLSX** reports using Python scripts (**openpyxl, python-docx**).
- Built centralized analytics dashboards using Excel, **Power Query, and Pivot Tables.**

Technical Projects & Innovations

1st Prize I-Novgames 2026 – Embedded AI & Anomaly Detection System
2026

Carelink Project – Anticipating et detecting epilepsy seizures

- **ECG signals analysis** from **PhysioNet (CHB-MIT) database**.
- Calcul d'indicateurs de variabilité cardiaque : **RMSSD, SDNN, LF/HF**.
- Trained **TensorFlow Lite models** on Edge Impulse; converted and **compressed model parameters for real-time execution on STM32** microcontrollers.
- Validated a proof-of-concept real-time alert system **across ~1,000 30-second windows (10 patients)** achieving **92% overall accuracy.**

Skin Maven Bandage
2025

Multi-Model Biometric AI Classification (Msc Project)

- Collected physical impedance dataset from **83 subjects**; applied **SMOTE augmentation** and class randomization**.**
- Data augmentation par **SMOTE** et randomisation.
- Benchmarked Machine Learning algorithms (**Logistic Regression, Random Forest, Gaussian Naive Bayes, KNN, SVC**).

Results: Random Forest: 90% accuracy on genre, 60% on age, 20% on skin tone.

Computer Vision Segmentation Pipeline on radiographic images
2025

- Implemented image processing pipelines in Python and Octave for brain and chest radiographic images (JPG format).
- Applied Otsu thresholding, color-space transformations, and morphological filtering (erosion/dilation) to extract target features.

Result: functional segmentation validated.